

ELECTIVE COURSE-7.5

UME751: LEAN MANUFACTURING				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: This course introduces the students to the complexity of production systems, dynamically changing product mix, short lead times and product life cycles, and the ever increasing pressures to contain costs. The course provides an understanding of the strategic role of operations management in effective planning for production to enhance firm's productivity and competitive advantage. The course also introduces some essential tools and techniques to move towards lean manufacturing. These include Just-in-time, Total Productive Maintenance, Single Minute Exchange of Die (SMED) etc. The course also provides exposure to the use of metrology related instruments/equipment utilized on the shop floor.				
Syllabus Introduction: Types of manufacturing systems: continuous and intermittent manufacturing; mass production, batch production and job shop production. Special cases under each production system. Need and strategy to reduce changeovers. Demand Forecasting: Need of lean manufacturing, uncertainty and volatility of demand, shortening product life cycles and rapid product introductions leading to the complexity of manufacturing systems, need of demand forecasting, cost of increased activity and opportunity cost, forecasting based on time horizon, forecasting based on methodology (time series analysis, econometric models, opinion based methods). Production Planning: Aggregate production planning, pure and mixed aggregate planning strategies; reactive aggregate planning strategies, aggressive aggregate planning strategies for lean manufacturing, Master Production Scheduling; Materials Requirements Planning, and Manufacturing Resource Planning. Lean Manufacturing Principles: The five interdependent principles of lean manufacturing- value definition, value stream analysis, just-in-time, flow and perfection; the seven different types of waste and their mitigation, lean manufacturing techniques- SMED, TPM, Just in Time etc.				
Laboratory Work • To obtain attribute data on a quality characteristic of given specimens to determine rejection levels using an Electrical Comparator (LVDT type comparator). • To obtain various dimensions (thickness, radius, angle etc.) on a given specimen using a Profile Projector. • To determine the angle on a given component using Sine Bar and Slip Gauges. • To determine the inside diameter of given specimens using (a) Jaw type Inside Micrometer, and (b) Rod type Inside Micrometer. • To determine the various angles on a single point cutting tool using a Tool Maker's Microscope. • To demonstrate the use of Vernier Bevel Protractor, Vernier Dial Caliper, Digital Caliper, Vernier Height Gauge, and Vernier Depth Gauge.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

- To demonstrate the use of Combination Set, Thread Micrometer, Radius Gauge, Feeler Gauge, and Dial Bore Gauge.
- To evaluate the surface roughness parameters of given specimens using a Surface Roughness Tester.

Course Learning Outcomes (CLOs)

The students will be able to:

1. select the most appropriate production system for a given manufacturing firm based on its product variety-production volume portfolio.
2. determine accurate forecasts for product demand using data on the past sales and the independent factors influencing the demand.
3. apply the procedures of manufacturing management to prepare aggregate plans, master production schedules, and materials requirements plans.
4. apply the principles and techniques of lean manufacturing to reduce waste and to improve the productivity of manufacturing systems

Text Books

1. Monks, J.G. *Operations Management, Second Edition, McGraw-Hills, (1996)*
2. Krajewski, L. J., Ritzman, L. P. and Malhotra, M. K., *Operations Management, Prentice Hall, New Delhi (2009).*

Reference Books

1. Chase, R. B., Aquilano, N. J. and Jacob, F. R., *Production and Operations Management: manufacturing and services, Tata McGraw Hill, New Delhi (1999).*
2. Christopher, M. *Logistics and Supply Chain Management, FT Prentice Hall, (2011).*

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1	MST	35
2	EST	35
3	Sessional (May include quizzes, lab evaluations etc.)	30

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